

Disproof of conjecture 00000000489:
 $S_3(n) = \sum_{\lambda \vdash n} f_\lambda^3/n!$ is at most $\sqrt{n!}$,
so it is not $\Theta(\sqrt{n!} n^{1/4})$

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1 The conjecture and the reading

Statement (English, verbatim). Definition: The third-order character sum $S_3(n) = \sum_{\lambda} f_\lambda^3/n!$. Conjecture: $S_3(n) = \Theta(\sqrt{n!} \cdot n^{1/4})$, and the growth of $S_3(n)/\sqrt{n!}$ is given by the main term of the 3-thread counting of the limit shape, with main-term constant $2^{1/4}\pi^{-1/2}$.

The Chinese text gives the same definition and, after a garbled prefix (“ $S_3(n) \sim c \cdot n^{1/4} \cdot (27/4)^{n^{2/4}+0+\dots}$ ”), says “precisely $S_3(n) = \Theta(\sqrt{n!} \cdot n^{1/4})$ ” with the same main-term constant $2^{1/4}\pi^{-1/2}$.

Objects. λ runs over the partitions of n , and f_λ is the dimension of the irreducible complex representation of the symmetric group S_n labelled by λ . The irreducible complex representations of S_n are, up to isomorphism, in bijection with the partitions of n (Section 4). Write $\sqrt{n!} = (n!)^{1/2}$, as in the statement’s own “ $S_3(n)/\sqrt{(n!)}$ ”, and let $n^{1/4}$ be the real positive fourth root.

Claims refuted. With $g(n) = \sqrt{n!} n^{1/4}$ and $C = 2^{1/4}/\sqrt{\pi}$:

(T) $S_3(n) = \Theta(g(n))$ as $n \rightarrow \infty$, i.e. $S_3 = O(g)$ and $g = O(S_3)$;

(L) $S_3(n)/g(n) \rightarrow C$;

(E) $S_3(n)/\sqrt{n!} \sim C n^{1/4}$.

(L) and (E) are our readings of “the growth of $S_3(n)/\sqrt{n!}$... with main-term constant $2^{1/4}\pi^{-1/2}$ ”. We do not give a meaning to “3-thread counting of the limit shape” and we do not address the garbled Chinese prefix beyond its explicit Θ claim. We prove $S_3(n) \leq \sqrt{n!}$ for every n , hence $S_3(n)/g(n) \leq n^{-1/4} \rightarrow 0$, which refutes the lower half of (T), and (L) and (E).

2 The mathematical proof

Let G be a finite group and $V_1, \dots, V_k$ pairwise non-isomorphic irreducible finite-dimensional complex representations with characters χ_i and dimensions d_i .

Lemma 1 ($\chi(g^{-1}) = \overline{\chi(g)}$). Let $A : G \rightarrow M_d(\mathbb{C})$ satisfy $A(gh) = A(g)A(h)$ and $A(1) = I$. Then $\text{tr } A(g^{-1}) = \text{tr } \overline{A(g)}$.

Proof. Put $P = \sum_{h \in G} A(h)^* A(h)$. Then $P = I + \sum_{h \neq 1} A(h)^* A(h)$ is positive definite, hence invertible. Reindexing $h \mapsto hg$ gives $A(g)^* P A(g) = P$. Since $A(g)A(g^{-1}) = I$, this gives $A(g)^* P = P A(g^{-1})$, so $A(g)^* = P A(g^{-1}) P^{-1}$ and $\text{tr } A(g) = \text{tr } A(g)^* = \text{tr } A(g^{-1})$. $\square$

Applying the lemma to the matrices of V in a basis gives $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Lemma 2. $\sum_i d_i^2 \leq |G|$.

Proof. By orthogonality of characters, $\sum_g \chi_i(g) \chi_j(g^{-1}) = |G| [i = j]$, because $V_i \cong V_j$ iff $i = j$. Let $\Phi = \sum_i d_i \chi_i$ and $D = \sum_i d_i^2$. Then $\sum_g \Phi(g) \Phi(g^{-1}) = |G| D$. By Lemma 1, $\Phi(g^{-1}) = \overline{\Phi(g)}$, so $|G| D = \sum_g |\Phi(g)|^2 \geq |\Phi(1)|^2 = D^2$, using $\chi_i(1) = d_i$. Hence $D \leq |G|$. $\square$

Theorem 3. $\sum_i d_i^3 \leq \sqrt{|G|} |G|$. For $G = S_n$ and $d_i = f_\lambda$ this is $S_3(n) \leq \sqrt{n!}$.

Proof. Each $d_i^2 \leq D \leq |G|$, so $d_i \leq \sqrt{|G|}$ and $\sum_i d_i^3 \leq \sqrt{|G|} \sum_i d_i^2 \leq \sqrt{|G|} |G|$. Divide by $|G| = n!$. $\square$

Hence $0 \leq S_3(n)/g(n) \leq n^{-1/4} \rightarrow 0$, so $S_3 = o(g)$. If $g = O(S_3)$ held, then $g = o(g)$, which is impossible since $g(n) \neq 0$ for $n \geq 1$. So (T) fails. (L) fails because the limit is $0 \neq C$. For (E): $(S_3(n)/\sqrt{n!})/(Cn^{1/4}) = (S_3(n)/g(n))/C \rightarrow 0 \neq 1$.

3 Lean formalization

The file `lean/Conjecture489/Basic.lean` (264 lines, `import Mathlib` only, Lean v4.33.1, Mathlib v4.33.1) proves the following in namespace `C489`.

- `trace_inv_eq_star` and `char_inv`: Lemma 1 and $\chi_V(g^{-1}) = \overline{\chi_V(g)}$ for every $V : \text{FDRep } \mathbb{C} \text{ } G$.
- `sum_sq_finrank_le`: for a finite type ι and $V : \iota \rightarrow \text{FDRep } \mathbb{C} \text{ } G$ with every V `Simple` and `Nonempty` ($V \text{ } i \cong V \text{ } j \rightarrow i = j$), $\sum_i (\dim V_i)^2 \leq |G|$. It uses `Mathlib's FDRep.char_orthonormal` and `FDRep.char_one`.
- `sum_cube_le`: $\sum_i (\dim V_i)^3 \leq \sqrt{|G|} |G|$.
- `IrrepFamily n`: a map $V : \text{Nat.Partition } n \rightarrow \text{FDRep } \mathbb{C} \text{ } (\text{Equiv.Perm } (\text{Fin } n))$, with each V_λ simple and distinct labels non-isomorphic. `IrrepFamily.f` is $f_\lambda = \dim_{\mathbb{C}} V_\lambda$ (`finrank`), and `IrrepFamily.S3` is $(\sum_\lambda f_\lambda^3)/n!$.
- `S3_le_sqrt`: $S_3(n) \leq \sqrt{n!}$ for every `IrrepFamily n`.
- `conjecture489_false`: for every $F : \forall n, \text{IrrepFamily } n$, (i) $S_3(n) \leq \sqrt{n!}$ for all n ; (ii) $S_3(n)/(\sqrt{n!} n^{1/4}) \rightarrow 0$; (iii) $S_3 = o(g)$ (`=o[atTop]`); (iv) $\neg S_3 = \Theta(g)$ (`=Θ[atTop]`); (v) the ratio does not tend to $2^{1/4}/\sqrt{\pi}$; (vi) $\neg (S_3(n)/\sqrt{n!}) \sim Cn^{1/4}$ (`~[atTop]`).

The theorem holds for every labelled family of pairwise non-isomorphic irreducibles. It does not assume that the family is complete, and it does not construct Specht modules. The family that defines the conjecture's f_λ (one irreducible for each partition, pairwise non-isomorphic) is one such family, so the conjecture's $S_3(n)$ is covered. Here $n^{1/4}$ is `Real.rpow`, and $0^{1/4} = 0$, so $g(0) = 0$; only $n \geq 1$ matters for the asymptotic claims.

Axiom audit. `lake build` succeeds, and `#print axioms` for `C489.conjecture489_false`, `C489.sum_sq_finrank_le` and `C489.char_inv` prints `[propext, Classical.choice, Quot.sound]`. The file contains no `sorry`, no `native_decide` and no added axioms.

4 Sources

Wikipedia, “Representation theory of the symmetric group” (https://en.wikipedia.org/wiki/Representation_theory_of_the_symmetric_group), retrieved 2026-10-05: “Therefore according to the representation theory of a finite group, the number of inequivalent irreducible representations, over the complex numbers, is equal to the number of partitions of n . Unlike the general situation for finite groups, there is in fact a natural way to parametrize irreducible representations by the same set that parametrizes conjugacy classes, namely by partitions of n or equivalently Young diagrams of size n .” The same article writes d_λ for “the dimension ... of the representation that corresponds to the Young diagram λ ”. No other source is used. All other steps are proved above and in Lean.